

Prithvi Sasikumar

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Summary

AI Engineer with **2+ years** of production experience developing and deploying scalable AI systems across Generative AI, RAG, Agentic AI, Computer Vision, and AI Automation. Delivered **73% model compression**, **sub-100 ms** real-time inference, **45%** reduction in manual inspection overhead, and enterprise-grade AI solutions leveraging LLMs, MLOps, AWS cloud infrastructure, and modern AI frameworks.

Professional Experience

AI Engineer Intern

OooWee LLC

June 2026 – Present

Remote

- Engineered **multi-agent AI pipelines** using **Gemini Pro** for automated lead qualification and client communication; designed multi-turn dialogue management with fallback routing logic, improving response consistency across production client accounts.
- Architected **LLM-powered automation workflows** integrating cross-platform APIs (CRM, project management, web) via function calling and tool orchestration; reduced manual handoff overhead across client-facing data pipelines through intelligent task routing.
- Applied advanced **prompt engineering** and **prompt chaining** techniques with **Gemini Pro** to automate business intelligence reporting, analytics synthesis, and research summarization; prototyped AI-driven solutions that cut recurring manual workload by **40%**.

Artificial Intelligence Engineer

Coherent Pixels Systems

December 2023 – July 2024

Chennai, India

- Architected and shipped **3** production AI systems for industrial automation, driving a **45%** reduction in manual inspection overhead; deployed end-to-end **PyTorch/TensorFlow** pipelines containerized with **Docker**, tracked via **MLflow**, and hosted on **AWS SageMaker**.
- Spearheaded a deep learning gait-analysis model using **HuggingFace Transformers** and transfer learning on 500+ annotated samples; applied Optuna hyperparameter tuning and Weights & Biases experiment tracking, boosting prediction consistency by **37%**.
- Delivered a real-time anomaly detection system at **<100 ms** latency via **TensorRT**-optimized **PyTorch** inference; automated model retraining with **CI/CD** on GitHub Actions and drift detection, cutting the false-negative rate by **52%**.

Project Experience

NexusAI: Multi-LLM Orchestration & RAG Platform — Live Demo: 13.58.124.199 (AWS EC2)

Personal Project

March 2026 – May 2026

Arizona, USA

- Architected and deployed a production-grade **Multi-LLM orchestration platform** on **AWS EC2** using Terraform IaC, Nginx reverse proxy, and **Docker Compose**; configured **GitHub Actions** CI/CD for automated zero-downtime deployments.
- Engineered an end-to-end **RAG** pipeline (**FastAPI**, **LangChain**, ChromaDB) with semantic chunking, hybrid vector retrieval, and semantic caching to reduce redundant LLM calls; built a Next.js **LLMOps** dashboard with real-time latency tracking, cache metrics, and SQLite-backed authentication.

ME-UP: RAG-Powered Hyperlocal AI Commerce Platform — github.com/Prithvi-01

Personal Project

December 2025 – March 2026

Arizona, USA

- Built a full-stack **Agentic AI** marketplace (**FastAPI**, **LangChain**, **GPT-4o**) with **Pinecone** vector search over MongoDB and Redis; achieved **sub-200 ms** end-to-end query latency via prompt engineering and semantic reranking for conversational product discovery.
- Engineered a **RAG** pipeline using **LangGraph** and ChromaDB for a buyer-seller chat system; integrated DALL-E image generation, fine-tuned LLM pricing suggestions, and a Streamlit dashboard — shipped with **Docker** and **GitHub Actions** CI/CD on **AWS S3**.

Real-Time Conveyor Defect Detection — Edge Machine Learning Inference Optimization — April 2024 – July 2024

github.com/Prithvi-01

Coherent Pixels Systems

Chennai, India

- Benchmarked **4** CNN architectures (MobileNetV2, ResNet, EfficientNet) for surface-defect classification; applied INT8 quantization via **ONNX** and **TensorRT**, compressing model size by **73%** and inference latency by **61%** while retaining **≥98%** accuracy.
- Tracked all experiments in Weights & Biases with FAISS-indexed feature stores on **AWS S3**; monitored production performance over a 3-month window with zero accuracy regression.

Technical Certifications

- Databricks:** Generative AI Engineer Associate | **Google:** Generative AI Fundamentals | **AWS:** Cloud Practitioner

Skills

Programming Languages: Python, C++, Java, SQL, Bash, MATLAB, R, JavaScript, TypeScript, HTML/CSS

Frameworks & Libraries: PyTorch, TensorFlow, Keras, FastAPI, Flask, LangChain, LangGraph, LlamaIndex, OpenCV, Hugging Face Transformers, Pandas, NumPy, scikit-learn, SciPy, Optuna, spaCy, NLTK, Matplotlib, PyTest

AI & Generative AI: Large Language Models, Retrieval-Augmented Generation, Agentic AI, AI Agents, Function Calling, Tool Calling, Prompt Engineering, Prompt Chaining, LLM Evaluation, Hallucination Mitigation, GPT-4o, Claude, Gemini, DALL-E, NotebookLM, Multimodal AI, Vision-Language Models, Conversational AI

Deep Learning & ML: Transformer Architecture, BERT, CNNs, Natural Language Processing, Named Entity Recognition, Object Detection, Anomaly Detection, Transfer Learning, Feature Engineering, Model Evaluation, Classification, Regression, Clustering, Time-Series Analysis, Computer Vision

MLOps & Deployment: Docker, AWS EC2, AWS S3, AWS Lambda, AWS SageMaker, Databricks, ONNX Runtime, MLflow, DVC, Git, Linux, GitHub Actions CI/CD, INT8 Quantization, TensorRT, Model Monitoring, Data Drift Detection, Automated Retraining, Model Serving

Data Stores & Vector DBs: PostgreSQL, MySQL, MongoDB, Redis, Elasticsearch, SQLite, Pinecone, FAISS, ChromaDB, pgvector, Weaviate, Qdrant, ETL Pipelines, Data Cleaning, Data Preprocessing, Metadata Stores, Vector Indexing, Similarity Search

Platforms & Tools: OpenAI API, Anthropic API, Google Gemini API, Hugging Face Hub, Weights & Biases, LangSmith, Make.com, GoHighLevel, CRM Automation, Workflow Automation, Trello, Wix, ROS2, NVIDIA Isaac Sim, Gazebo, RViz, Streamlit, Jupyter, VS Code, Jira, GitHub

Education

M.S. Robotics and Autonomous Systems (Artificial Intelligence)

Arizona State University, Tempe, AZ

August 2024 – May 2026

CGPA: 3.87/4

Bachelor of Engineering in Computer Science

Easwari Engineering College, Chennai, India

August 2020 – May 2024

CGPA: 3.33/4